INTRODUCTION TO INFORMATION GEOMETRY

Potentials, Stability, and KL Flows

Week 9 · Lecture 2

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When a payoff matrix has a potential

Suppose fitness is $f(p) = Mp$, with M symmetric. Define

$$\Phi(p) = \tfrac{1}{2} p^T M p.$$

Then $\partial_i \Phi = (Mp)_i = f_i$, and the replicator equation is the Fisher gradient of Φ.

An example that favors balance

Choose $M = -I$. Each strategy's fitness is $f_i = -p_i$, and

$$\Phi(p) = -\frac{1}{2}\sum_i p_i^2.$$

For a fixed total probability, the sum of squares is smallest at the uniform distribution.

Thus the uniform state maximizes this potential.

Compute a balancing flow

For $M = -I$, calculate the replicator velocity at

$$P = (.6, .3, .1).$$

Use $f_i = -p_i$ and $p_i' = p_i(f_i - \bar{f})$.

Compute a balancing flow

For $P = (.6, .3, .1)$, the mean fitness is $\bar{f} = -\sum p_i^2 = -.46$. Therefore

$$p' = (-.084,\ .048,\ .036).$$

The largest probability falls; the smaller two rise.

Verify ascent at the same state

At $P = (.6, .3, .1)$, we found $p' = (-.084, .048, .036)$. Evaluate the potential derivative using

$$\dot{\Phi} = \sum_i f_i p_i' = -\sum_i p_i p_i'.$$

Verify ascent at the same state

At $P = (.6, .3, .1)$,

$$\dot{\Phi} = -(.6)(-.084) - (.3)(.048) - (.1)(.036).$$

Thus

$$\dot{\Phi} = .0324 > 0.$$

Curvature of the potential and stability

At an interior critical point q, the derivative of a metric gradient field is

$$DF(q) = G(q)^{-1} \operatorname{Hess} \Phi(q).$$

If the Hessian is negative definite on the tangent space, then q is locally attracting.

Why gradient equilibria have real eigenvalues

The linearization $G^{-1}H$ need not itself be symmetric. But it is similar to

$$G^{-1/2}HG^{-1/2},$$

which is symmetric whenever $H = \operatorname{Hess}\Phi$ is symmetric. Consequently its eigenvalues are real.

A target distribution as a potential

Let $Q = (q_1, \ldots, q_n)$ have strictly positive coordinates. Define

$$\Phi(P) = -D(Q\|P) = -\sum_i q_i \ln \frac{q_i}{p_i}.$$

The potential is largest when $P = Q$. What does its Fisher-gradient flow look like?

The target-seeking flow is linear

Differentiate the potential with respect to p_i:

$$\frac{\partial \Phi}{\partial p_i} = \frac{q_i}{p_i}, \qquad \sum_i p_i \frac{\partial \Phi}{\partial p_i} = 1.$$

The natural-gradient equation simplifies to

$$p_i' = q_i - p_i.$$

Every coordinate relaxes toward its specified target.

Solve the target-seeking system

Consider $P' = Q - P$, where Q is fixed. Solve the initial-value problem $P(0) = P_0$. What happens to the distance $P(t) - Q$ as time increases?

Solve the target-seeking system

Consider $P' = Q - P$, with $P(0) = P_0$. Each coordinate satisfies a first-order linear equation:

$$P(t) = Q + e^{-t}(P_0 - Q).$$

The displacement from the target decays by e^{-t}.

One step toward a chosen target

Let

$$Q = (.3, .4, .3), \qquad P_0 = (.65, .2, .15).$$

Use $P(1) = Q + e^{-1}(P_0 - Q)$. Find $P(1)$ to three decimal places.

One step toward a chosen target

Let $Q = (.3, .4, .3)$ and $P_0 = (.65, .2, .15)$.

$$P(1) = (.3 + .35/e,\ \ .4 - .2/e,\ \ .3 - .15/e).$$

Thus

$$P(1) \approx (.429, .326, .245).$$

Reverse the KL divergence

Now choose the distinct potential

$$\Phi(P) = -D(P\|Q) = -\sum_i p_i \ln \frac{p_i}{q_i}.$$

Its Fisher gradient is

$$p_i' = p_i \left[-\ln \frac{p_i}{q_i} + D(P\|Q) \right].$$

Reverse-KL flow has a geometric interpolation

Let $a(t) = e^{-t}$. The reverse-KL solution is

$$p_i(t) = \frac{q_i^{1-a(t)} p_i(0)^{a(t)}}{\sum_j q_j^{1-a(t)} p_j(0)^{a(t)}}.$$

This path interpolates multiplicatively between the initial distribution and the target, then renormalizes.

Reverse-KL flow toward uniformity

Suppose $Q = (1/3, 1/3, 1/3)$ and $P_0 = (.64, .25, .11)$. At $t = \ln 2$, find the unnormalized weights $p_i(0)^{e^{-t}}$. How would you turn them into a probability distribution?

Reverse-KL flow toward uniformity

Suppose $Q = (1/3, 1/3, 1/3)$ and $P_0 = (.64, .25, .11)$. At $t = \ln 2$, $e^{-t} = 1/2$, so the weights are

$$(.8, .5, \sqrt{.11}).$$

Dividing by $.8 + .5 + \sqrt{.11}$ gives $P(t) \approx (.490, .306, .203)$.

Why every flow is not a gradient flow

An increasing potential cannot return to its starting value along a nonstationary closed orbit. The balanced three-strategy cyclic replicator has closed orbits. Such cyclic motion

cannot be globally realized as ascent of a single smooth Fisher potential.

Dynamics constrain the possible geometry

When fitness comes from a potential, its Fisher-gradient flow provides monotone ascent and a direct stability test. When motion cycles, a single global ascent potential is unavailable. These two behaviors give us complementary ways to organize the dynamics of probability distributions.